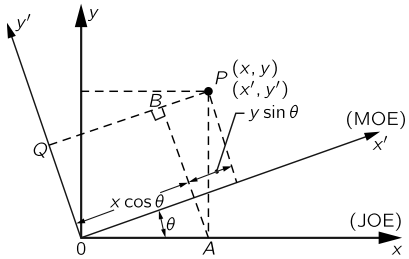

Figure 1-2 Two coordinate systems having different angular orientations.

lengths along the x'-axis, and y' as the difference of two lengths along AB. All these lengths are expressed in terms of x, y, and θ in equations (1.5), to which we have added an equation for the third dimension.

$$\begin{aligned} x' &= x \cos\theta + y \sin\theta, \\ y' &= y \cos\theta - x \sin\theta, \\ z' &= z. \end{aligned} \tag{1.5}$$

The next step is to analyze the relationship of forces as seen by the two observers, following the same general method as before. Let us assume that a force F, which has already been analyzed as having components F_x and F_y (as seen by Joe), is acting on a particle of mass m, located at point P in Figure 1-2. For simplicity, let us move both sets of axes so that the origin is at P, as shown in Figure 1-3. Moe sees the components of F along his axes as $F_{x'}$, and $F_{y'}$. F_x has components along both the x'- and y'-axes, and F_y likewise has components along both these axes. To express $F_{x'}$, in terms of F_x and F_y, we sum these components along the x'-axis, and in a like manner we can express $F_{y'}$ in terms of F_x and F_y. The results are

$$\begin{aligned} F_{x'} &= F_x \cos\theta + F_y \sin\theta, \\ F_{y'} &= F_y \cos\theta - F_x \sin\theta, \\ F_{z'} &= F_z. \end{aligned} \tag{1.6}$$